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CFD Analysis of Thermal-Hydraulic Behavior of PCCS

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Abstract

Today, Computational Fluid Dynamics is an indispensable tool for the simulation and analysis of complex phenomena. It is used in research to accurately model physical interactions that are difficult to reproduce experimentally. In this study, we carried out a double-precision 3D CFD simulation of a counter current heat exchanger integrated into the passive containment cooling system (PCCS). The aim is to analyze condensation heat transfer in the presence of non-condensable gases (NCG), a complex phenomenon that is difficult to model using conventional approaches. In this work, the numerical prediction is based on a forced convection model using the RNG k-epsilon turbulence model. The exchanger geometry is based on Kuhn's experimental study, enabling us to validate our results by direct comparison. Several tests were carried out to assess the model's consistency and to define its range of validity. The results show that CFD provides a good prediction of the heat transfer coefficient (HTC) and a better understanding of the effects of non-condensable gases, which slow down condensation and form thermal barriers. Compared with tools such as RELAP5, CFD offers a better representation of local phenomena, with fewer approximations.

Key-words : Computational Fluid Dynamics (CFD), Numerical Simulation, Condensation Heat Transfer, Non-condensable Gases,Passive ontainment Cooling Systems (PCCS), Nuclear Safety, Thermo-hydraulic Modeling

Methodology

The numerical methodology used to study flow and heat transfer in a counter-current heat exchanger relies on solving the Reynolds-Averaged Navier-Stokes (RANS) equations to model turbulent flow. Simulations were carried out using ANSYS Fluent software, which solves the following equations:

- Continuity equation:

$$\frac{\partial \overline{U}_i}{\partial x_i} = 0$$

- Momentum equation :

$$\frac{\partial \overline{U}_i}{\partial t} + \overline{U}_j \frac{\partial \overline{U}_i}{\partial x_j} = \frac{-1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\mu}{\rho}\frac{\partial^2 \overline{U}_i}{\partial x_j \partial x_j} - \frac{\partial (u'_i u'_j)}{\partial x_j}$$

- Energy equation:

$$\rho Cp\left(\frac{\partial \overline{T}}{\partial t} + \overline{U}_j \frac{\partial \overline{T}}{\partial x_j}\right) = \frac{\partial}{\partial x_j}\left(k\frac{\partial \overline{T}}{\partial x_j} - \rho Cp u'_j T'\right)$$

- The condensation rate is governing by the rate of diffusion of water vapor toward the cold surface, Bird formulation is used for each species:

  For the NCG : $\dot{m}''_{nc} = \mu W_{nc} - \rho D \frac{\partial W_{nc}}{\partial n}$

  For the steam : $\dot{m}''_{s} = \mu W_{s} - \rho D \frac{\partial W_{s}}{\partial n}$

- The RNG k-epsilon turbulence model was used for turbulence modeling.
- To optimize the CFD simulation, only a 15/360 portion of the exchanger was modeled in 3D, taking advantage of the system's cylindrical symmetry.

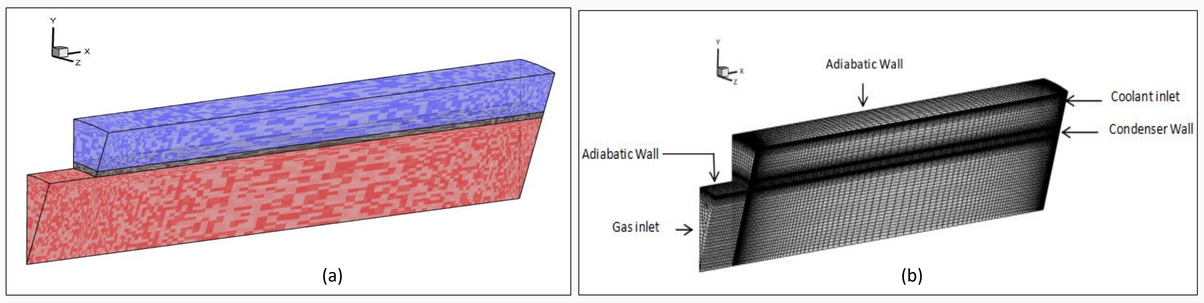

Figure 1: Image of the domain geometry under study, (a) represents the flow of hot and cold fluids, (b) the domain mesh.

Nodes	Cells	Faces
94078	80420	254953

Table 1: Mesh properties.

As the geometry was identical to that used by Kuhn [3], a validation was carried out by comparing the simulation results with the experimental data from his work, presented in the figure below.

Results :

Cases	Pressure (Pa)	Discrepancy
Run 2.1-13	412600	6.2%
Run 2.1-13R	420100	0.38%
Run 2.2-4	116300	7.7%
Run 2.2-8	115700	1.25%
Run 2.2-13	120500	2.25%

Table 2: relative errors between experimental and CFD results

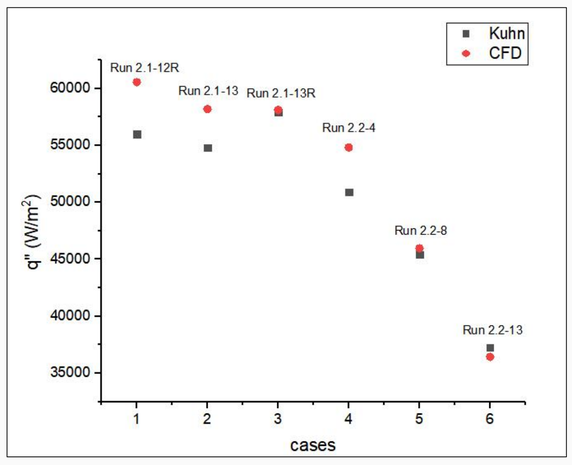

Figure 2: Comparison between Kuhn's experimental values and CFD results

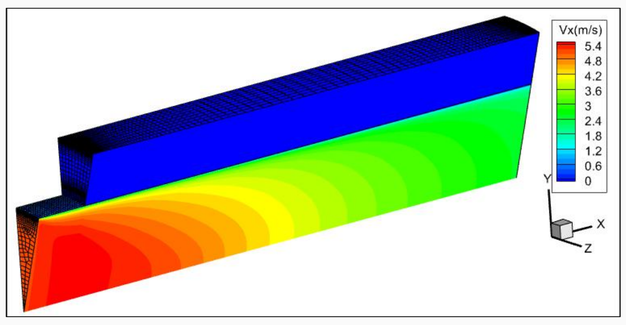

Figure 3: Axial velocity contour

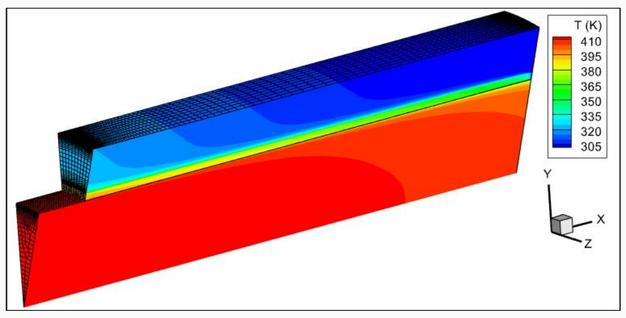

Figure 4: Contours of temperature

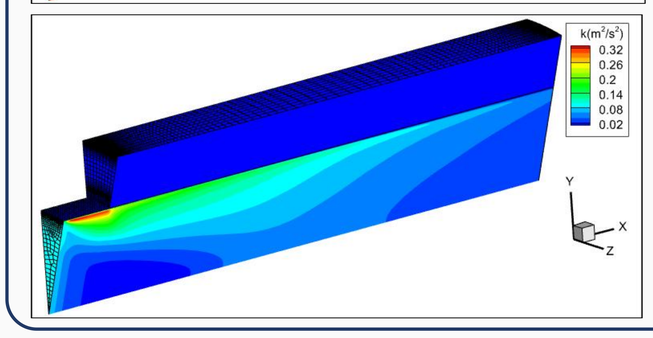

Figure 5: Contours of tturbulent kinetic energy.

Conclusion

This study shows that the CFD approach is a relevant alternative to experimentation for analyzing heat transfer in PCCS. It offers time and cost savings while allowing the exploration of complex conditions, which is especially valuable in contexts where experimental data is limited. Moreover, the simulation results were found to be acceptable and relatively close to experimental data, confirming the reliability of this method for predicting the thermal behavior of passive cooling systems.

Références

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[3] Shine-Zen Kuhn, Investigation of Heat Transfer from Condensing Steam-Gas Mixtures and Turbulent Films Flowing Downward Inside a Vertical Tube, PhD, University of California at Berkeley, USA, 1995